Supply Chain Analytics 1 (2023) 100004

Contents lists available at ScienceDirect

Supply Chain Analytics

journal homepage: www.sciencedirect.com/journal/supply-chain-analytics

A two-stage data envelopment model for evaluating the exchange efficiency of the imports and exports in a railway supply chain network

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ARTICLE INFO

Keywords:
Data envelopment analysis
Two-stage model
Railway supply chain network
Exchange efficiency
China railway

ABSTRACT

This study presents a Data Envelopment Analysis (DEA) model to evaluate the efficiency of the exchange of regional import and export goods in a railway supply chain network. Unlike previous railway supply chain efficiency studies, the exchange efficiency incorporates railway supply chain operational input, import volume as input, and export volume as output to help policymakers review railway supply chain performance and local freight structure decisions. The transport service efficiency of transportation lines between provinces in the railway supply chain network is formulated with a two-stage DEA model. This transport service efficiency is further decomposed into the efficiency of the forward and reverse lines constituting the transportation lines between provinces. To provide decision-makers with a path to improve exchange efficiency, we measure the improvement potential for each input and output of a province. The empirical study of China's railway supply chain shows (1) Shanxi, Tibet, Shanghai, and Hainan provinces have the highest exchange efficiency; (2) intra-regional transport service efficiency is higher than inter-regional transport service efficiency in North, East, and Southwestern China; and (3) provinces with lower exchange efficiency can also have transportation lines with high transport service efficiency.

1. Introduction

In the past 20 years, China has developed its railway supply chain more rapidly than any other Asian country. At the beginning of the 21st century, the mileage of China's national railways in operation was 68,700 km, with a passenger turnover of 453.26 billion passenger-km and cargo turnover of 1377.05 billion ton-km. But by 2018, the mileage of China's national railways in operation was 139,900 km, with a passenger turnover of 1414.66 billion passenger-km and cargo turnover of 2882.06 billion ton-km. Meanwhile, the coverage of China's railway supply chain network has been greatly improved. For example, the central government completed several railway projects in its *11th five-year plan of western region development* such as the Qinghai-Tibet Railway. In 2013, China announced the 'One Belt And One Road' plan to realize the common economic development of the nearly 70 countries locating in most of Asia, South America, coastal areas (Indian Ocean coast and Mediterranean coast). Railway transportation is one of the key factors driving the rapid economic development along the Belt and Road. To this end, China has developed a railway transport brand called "China Railway Express", which has successfully connected 17 countries such as Russia, Poland, Belgium, Germany, and Britain. In 2014, China had the second largest railway supply chain network in the world, according to the *Bulletin of railway statistics*, with railway transporting having become one of the main modes of transportation for inter-provincial goods exchange in China.

In the performance evaluation of public transportation, DEA is a widely used non-parametric tool to compare the efficiency of homogeneous decision-making units [7]. Many studies used DEA to study railway supply performance in order to provide policy suggestions for decision-makers to further improve railway supply chain efficiency. However, few studies have examined the efficiency of regions (provinces in the present study) in importing and exporting goods by railway supply chain. As a transport medium of goods exchange brought by trade, railway transportation can reflect the railway economic structure of a certain region and the railway economic relations between different regions. When evaluating the exchange efficiency of railway supply chain, the input-output of this paper includes the operational input of railway supply chain, such as staff, length of transport

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https://doi.org/10.1016/j.sca.2023.100004
Received 11 February 2023; Received in revised form 3 March 2023; Accepted 6 March 2023
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route, as well as the inflow and outflow of volumes. Thus, the evaluation of goods exchange efficiency by railway supply chain between regions can not only provide the government with suggestions to improve the railway supply chain efficiency but also help the government to review and/or rearrange the freight structure of railway trade to improve the regional economy. The contribution of this paper is as follows:

First, a DEA-based model is developed to measure and compare the exchange efficiency of 31 Chinese mainland provinces. This exchange efficiency model reflects the structure of good exchange in the railway supply chain network thereby accounting for the volume of import and export of railway goods between provinces and railway production factors. It is shown that an analysis of the exchange efficiency provides helpful insights for decision-makers to improve the railway operations involving goods transportation and exchange. To the best of our knowledge, this exchange efficiency model is a novel application of DEA in transportation.

Then, we open the black-box of exchange efficiency to define the transport service efficiency of transportation lines between provinces in the railway supply chain network by a two-stage DEA approach. The notion is that an evaluation of the transport service efficiency for the multiple transportation lines between provinces provides managers and policy makers with detailed information on what causes the exchange efficiency of the provinces in the railway supply chain network. Using this detailed information, policy makers and railway management can identify less efficient transportation lines in the railway supply chain network and implement measures to improve the transport service efficiency of these lines (and, by doing so, improve exchange efficiency of the provinces in the network as well as the global efficiency in the railway supply chain network). Given that a railway line generally allows two-way transport and exchange of goods, in the model a transportation line between two provinces is modelled as a combination of a forward line and a reverse line. By modeling the transportation lines as two-way traffic and exchange lines, the results of the transport service efficiency model provide insights into the transport service efficiency of a transportation line as well as the efficiency of the composing forward and reverse lines, so as to formulate a more detailed improvement plan.

Furthermore, using the multi-directional efficiency approach (MDEA) as introduced by Asmild et al. [1], we measure the improvement potential for the different railway inputs and outputs of the provinces and provide for each province a path for increasing the exchange efficiency. In this paper, the proposed models are applied to railway transportation of 31 Chinese provinces in 2021 to analyze the provincial railway supply chain exchange efficiency from the macro perspective and compare the inter-provincial transport service efficiency from the micro perspective, providing multi-level policy suggestions for the government.

The rest of this paper is structured as follows: Section 2 summarizes the DEA-literature on efficiency analysis in the railway supply chain. The exchange efficiency model, transport service efficiency model, and potential efficiency improvement analysis are introduced in Section 3. Section 4 shows the empirical study of Chinese railway supply chain based on the proposed models. Section 5 summarizes the conclusion and shows the prospect.

2. Literature review

Most studies in the DEA literature measure the efficiency performance of the railway supply chain. For example, in evaluating the performance of Indian Railways, Bhatia and Sharma [6] tested the robustness of the results using CRS (Constant Return to Scale) and VRS (Variable Return to Scale) models respectively. The results show that technological innovation and employee management are the most important factors to reduce costs. The traditional DEA model can only deal with quantitative indicators when evaluating railway performance. In order to make up for this shortcoming, da Silva et al. [11] focused on how to improve the safety of the rail system. Based on DEA and Monte Carlo methods, they proposed an evaluation method to identify defects in railway castings. Marchetti and Wanke [30] assessed the efficiency performance of Brazil's local railways after the government issued new transport regulations. They found that the volume of broad-gauge bulk commodities has an important impact on local railway supply chain efficiency. Kwon [21] combined artificial neural networks and the DEA-method to evaluate and predict the efficiency performance of American railroads. Hsiao et al. [19] examined the potential of reallocating resources to improve railway transportation efficiency in Taiwan. To provide policymakers with better advice, they assessed the performance change of the Taiwan railway bureau with different resource allocation plans. Djordjević et al. [13] developed a non-radial DEA-approach to evaluate the safety of railway level crossings (intersection where railway meet road) safety, taking into account undesirable outputs. Chiou et al. [9] designed a route-based DEA-approach to not only measure 37 bus operators' performance but also gave the efficiency of all routes under each operator.

Some papers used the DEA-method to measure the efficiency and effectiveness of transportation services [12,16,23,48,49]. Tavassoli and Saen [39] put forward a fuzzy DEA model which can deal with both qualitative and quantitative indicators to evaluate the environmental efficiency of the railway supply chain. Graham [17] used both parametric (total factor productivity) and non-parametric (DEA) methods to calculate the operational efficiency of 200 railway supply chain around the world and found that the two methods produced largely the same efficiency ranking. Based on the observation that railway data are sometimes not very accurate, [44] used a fuzzy DEA-method to evaluate and rank the railway supply chain efficiency of Asian countries. They found that the inaccuracy of the data and the model choice may have a considerable impact on the efficiency scores and ranks.

Several DEA-papers have also focused on measuring the environmental efficiency in railway supply chain [47]. For example, Michali et al. [31] evaluated the environmental efficiency of railway supply chains in 22 European countries based on the network DEA. In particular, they take into account noise pollution from the railway supply chain when assessing environmental efficiency. Liu et al. [25] built up a Slack-Based DEA-method to assess the environmental efficiency of China's land transportation sector thereby accounting for the railway and highway supply chain as sub-sectors in parallel structures. Song et al. [38] used a two-stage DEA-process to study analysis factors affecting the environmental efficiency of the railway supply chain in China. They showed that the environmental efficiency in railway supply chain steadily increased from 2006 to 2011 in China. Oum et al. [33] established a novel directional distance function to assess the performance of transportation modes (railroad and airline) when passengers traveling between cities in Japan. A key finding was that railroad has higher social efficiency than airline. Cui and Li [10] introduced a virtual frontier DEA-model into a three-stage DEA structure for the first time to calculate transportation energy efficiencies of the 30 provinces in China. In doing so, they defined the concept of transportation energy efficiency as the ratio of transport output (passenger and freight turnover volume) to input (labor, capital and energy). Lv et al. [28] extended this concept of transportation energy efficiency by also accounting for the air pollutants and spatial spillover effects air pollutants on transportation energy efficiency. In a study of container multimodal transport in Chinese ports, Wang et al. [43] found that less use of railway transport was the main reason for higher carbon dioxide production in container transport.

Most of the above articles used DEA-models to measure and evaluate railway supply chain performance, without considering the internal activities of the railway system (i.e., the efficiency of the cargo transport between two regions/provinces in the railway supply chain network). Only few papers attempted to open the black-box of railway supply chain efficiency. Since Taiwan Railway suffered from perennial losses, Hsiao and Chen [18] proposed a three-stage method based on

the DEA model to reallocate railway resources to improve the performance of Taiwan Railway. Liu et al. [26] evaluated the provincial energy-environment efficiency of railway supply chain using a two-stage process. In the first stage, they introduced an undesirable output (carbon dioxide) to evaluate the efficiency of the railway supply chain based on a non-radial DEA-approach. In the second stage, they selected income, investment, population density, and other exogenous factors to investigate the main influencing factors of the efficiency via a Tobit regression. The results demonstrated, amongst all regions, that western China's railway supply chain is the most efficient. Wanke et al. [45] decomposed railway supply chain efficiency into operational efficiency and revenue efficiency, with the two sub-efficiencies being connected through common intermediate products (i.e., cargo turnover and passenger turnover). Their results showed that China's railway supply chain had high efficiency in freight transportation, while the railway supply chain of other countries like Japan, had high passenger transportation efficiency.

Some studies examined the efficiency of the railway supply chain thereby accounting for the complex nature of the railway supply chain network (i.e., with the network connecting multiple provinces). For instance, combined with DEA and Tobit regression, Le et al. [22] evaluated and analyzed the performance and influencing factors of 18 Tokyo railway operating lines. Mohammad Hasany and Shafahi [32] used a two-stages stochastic program to study the railway supply chain congestion caused by uncertain supply and demand between two provinces in the railway supply chain network. Qian et al. [34] analyzed the influence of the proportion of various transportation modes (i.e., air, waterway, road, and railway) on the transportation efficiency of stations in a network. Rotoli et al. [36] used a DEA-model to assess the accessibility of rail systems across countries (NUTS 3-level) in Europe and identified infrastructure development and speed improvements as ways to continue to improve rail accessibility. Bai et al. [5] used the resampling DEA-approach to discuss what would be the changes in local railway supply chain efficiency caused by mergers and acquisitions of railway authorities in the railway supply chain network.

Some papers used DEA-type models to examine the potential of different policy and industry measures (i.e., government subsidies, cost control, etc.) for improving the efficiency of the railway supply chain. For instance, Tomikawa and Goto [41] compared changes in railway supply chain performance before and after privatization in Japan. They found that privatization helped improve the Japanese railway's operational efficiency. Mallikarjun et al. [29] combined an unoriented DEA and Network DEA model to assess the organizational performance of the railway supply chain in the U.S. using American railway data from 2001 to 2010. Using a two-stage DEA-model, Link [24] investigated the efficiency of German public transport authorities (PTAs) in the use of available public funding, thereby accounting for railway service quality in the efficiency measurement. She found that PTAs with longer and smaller contracts, and a higher share of tendering and gross contracts were more efficient than other PTAs, even more so when accounting for service quality in the efficiency measurement. Roets et al. [35] proposed a multi-output DEA-method with cost allocation restrictions to measure and monitor staff efficiency of railway traffic control staff working at Belgian computerized railway traffic control centers and detect operational safety concerns. Rotoli et al. [37] used a DEA-method to study the performance changes of railway operators under different pricing systems before and after the promulgation of the new railway regulations (market opening) by the Italian government and provided useful reference opinions for the authorities. Topcu et al. [40] used the DEA-method to take the Belgium railway supply chain company as an example to discuss the impact of the working environment and workload of railway personnel on company performance. More research could be found in Jose et al. [20], Dolinayova and Domeny [14], Xu and Wang [46].

To sum up, although there have been several DEA-based studies on railway supply chain efficiency, less attention has been paid to exchange efficiency of railway supply chain. This paper aims to construct the DEA-framework for evaluating the exchange efficiency of (regions/provinces in) the railway supply chain network in order to provide helpful insights and suggestions on how to improve railway supply chain efficiency for decision-makers and stakeholders in the railway sector.

3. Methodology

3.1. Provincial exchange efficiency

In the present setting, a railway supply chain network is defined by the frequent trade and exchange of goods between provinces through rail transport. The flow of goods within the provinces will not be considered in this paper. In a multi-product railway supply chain economy, goods produced in different provinces are redistributed by rail according to market demand. The goods exported by one province may be produced using the natural resources of its own province and the raw materials of other provinces or may be imported from other provinces and exported after processing. Therefore, the traffic outflow volume of railway goods in a province is directly related to the daily operation capacity of the railway supply chain and the traffic inflow volumes into the province. This paper defines a new concept, namely exchange efficiency, to discuss the relationship among the three by DEA method. Without the loss of generality, suppose there are n provinces and p ($p = 1, \ldots, P$) kinds of goods in the railway supply chain network N, each province j ($j = 1, \ldots, n; \forall j \in N$) has daily operational inputs $X_j = (x_{1,j}, \ldots x_{B,j})$; traffic inflow volumes $Y_j^{input} = \{(y_{1,j}^{input}, \ldots, y_{P,j}^{input}) | y_{p,j}^{input} = \sum_{k \in N \setminus \{j\}} y_{p,k}^{input}, \forall p\}$; traffic outflow volumes $Y_j^{output} = \{(y_{1,j}^{output}, \ldots, y_{P,j}^{output}) | y_{p,j}^{output} = \sum_{k \in N \setminus \{j\}} y_{p,k}^{output}, \forall p\}$.

Fig. 1 shows an example of a small railway supply chain network consisting of three provinces: Province A, Province B, and Province C. All three provinces are connected by rail and there is movement of goods between them. In the small railway supply chain network, each of the provinces uses daily operational inputs (i.e., staff, length of transport route, etc.) to organize the import of goods from and the export of goods to the other provinces in the network. For example, Province A has a railway input (X_A) to organize the import of goods (Y_{BA}, Y_{CA}) from and exports of goods (Y_{AB}, Y_{AC}) by rail to the Province B and Province C, respectively. In the railway supply chain network, the exchange efficiency of goods transported at the evaluated province is calculated as in Fig. 2:

Formally, the exchange efficiency of the evaluated province is computed by model (1):

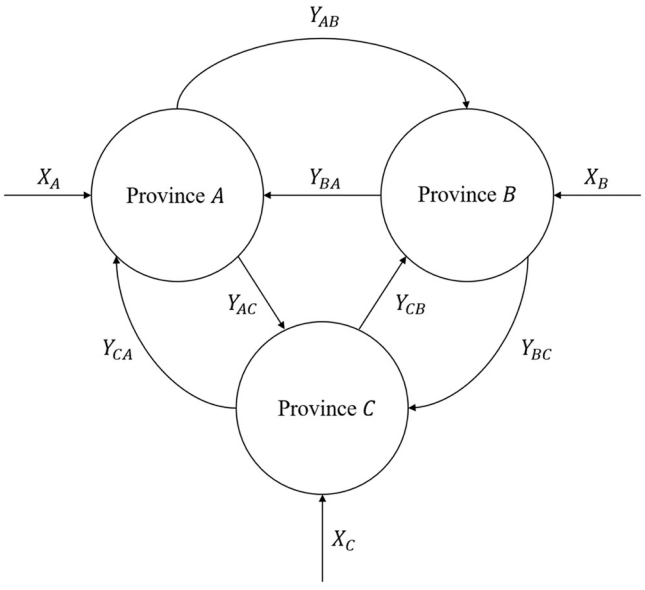


Fig. 1. A small railway supply chain network.

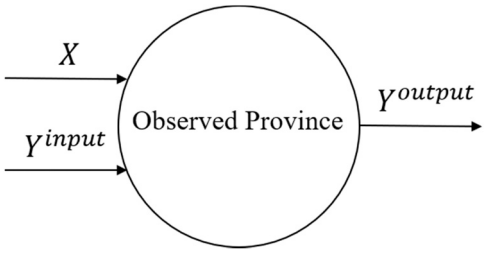


Fig. 2. Exchange efficiency of the evaluated province.

$$e_0 = \max \frac{U^T Y_0^{output} - u_0}{V^T(X_0,\ Y_0^{input})}$$

$$s.\ t. \begin{cases} \dfrac{U^T Y_j^{output} - u_0}{V^T\left(X_j,\ Y_j^{input}\right)} \le 1 & \forall\ j \quad \text{(i)} \\ U^T,\ V^T \ge \varepsilon & \text{(ii)} \\ u_0 \quad \textit{free in sign} & \text{(iii)} \end{cases} \tag{1}$$

where U^T and V^T (T stands for transpose) are the weight vectors of corresponding inputs and outputs; u_0 is the unrestricted variable for returns to scale technology (VRS, see [4]); ε is the infinitesimal value (10^{-6}) to require U^T and V^T greater than zero to ensure the effectiveness of inputs and outputs. By letting $T = \frac{1}{V^T(X_0,\ Y_0^{input})}$, model (1) is transferred into the following linear programming (via Charnes and Cooper's [8] transformation):

$$e_0 = \max(U^T Y_0^{output} - u_0)$$

$$s.\ t. \begin{cases} U^T Y_j^{output} - V^T(X_j,\ Y_j^{input}) - u_0 \le 0 & \forall\ j \quad \text{(i)} \\ V^T(X_0,\ Y_0^{input}) = 1 & \text{(ii)} \\ U^T,\ V^T \ge \varepsilon & \text{(iii)} \\ u_0 \quad \textit{free in sign} & \text{(iv)} \end{cases} \tag{2}$$

The dual programing of the model (2) is expressed as:

$$e_0 = \min\left[\theta - \varepsilon\left(\sum\nolimits_p s_p^{output+} + \sum\nolimits_b s_b + \sum\nolimits_p s_p^{input+}\right)\right]$$

$$s.\ t. \begin{cases} \sum_{j=1}^n \lambda_j y_{p,j}^{output} - s_p^{output} = y_{p,0}^{output} & \forall\ p \quad \text{(i)} \\ \sum_{j=1}^n \lambda_j x_{b,j} + s_b = \theta x_{b,0} & \forall\ b \quad \text{(ii)} \\ \sum_{j=1}^n \lambda_j y_{p,j}^{input} + s_p^{input} = \theta y_{p,0}^{input} & \forall\ p \quad \text{(iii)} \\ \sum_{j=1}^n \lambda_j = 1 & \text{(iv)} \\ \lambda_j,\ s_p^{output},\ s_b,\ s_p^{input} \ge 0 & \forall\ j,\ p,\ b \quad \text{(v)} \end{cases} \tag{3}$$

3.2. Transport service efficiency between two provinces

The model established in the previous section evaluates the exchange efficiency of the provinces in the railway supply chain network from a macroscopic perspective. To assess the efficiency of the transportation lines between the provinces in the railway supply chain network, the black-box of model (1) is opened using a two-stage DEA-method. Fig. 3. illustrates the general transport structure of a railway line between two provinces, i.e., province i and province j, in the railway supply chain network.

The above two-stage transport structure reflects the flow of goods on the railway transportation lines between province i and province j. Suppose that the set S represents all the transportation lines in the railway supply chain network. For an observed transportation line $s_0 (s_0 \in S)$ formed by the provinces i and j in the railway supply chain network as shown in Fig. 3, $X_{s_0}^i = (x_{1,s_0}^i, \ldots x_{B,s_0}^i)$ are inputs to province i (inputs that are shared among the multiple transportation lines connecting province i to province j and the other provinces in the railway supply chain network); $Y_{s_0}^{input-i} = \{(y_{1,s_0}^{input-i},\ \ldots, y_{P,s_0}^{input-i}) |\ y_{p,s_0}^{input-i} = \sum_{k\in S\setminus\{s_0\}} y_{p,k}^{input-i},\ \forall\ p\}$ the total amount of goods imported into province i from the other provinces in the network N (except from province j) via transportation lines other than s_0; $Y_{s_0}^{output-i} = \{(y_{1,s_0}^{output-i},\ \ldots, y_{P,s_0}^{output-i}) |\ y_{p,s_0}^{output} = \sum_{k\in S\setminus\{s_0\}} y_{p,k}^{output-i},\ \forall\ p\}$ the total amount of goods exported from province i to the other provinces in the railway supply chain network N (except to province j) via transportation lines other than s_0; $F_{s_0} = (F_{1,s_0},\ \ldots, F_{P,s_0})$ the goods exported from province i to province j using the transportation line s_0 (can be regarded as the freight transport on the forward line of the transportation line s_0); $R_{s_0} = (R_{1,s_0},\ \ldots, R_{P,s_0})$ the goods imported from province j in province i (can be regarded as the freight transport on the reverse line of the transportation line s_0). Similarly, we can define for province j the following inputs and outputs: $X_{s_0}^j$, $Y_{s_0}^{input-j}$, and $Y_{s_0}^{output-j}$. The two-stage structure in Fig. 3 includes not only the goods exchange between two provinces i and j (freight transport on the forward line and reverse line of the transportation line s_0), but also the goods exchange between these two provinces and the rest of the provinces in the railway supply chain network. To measure the transport service efficiency at the transportation line s_0 that consists of a forward transportation line and reverse transportation line, we first calculate the overall transport service efficiency of the transport structure shown in Fig. 3 as:

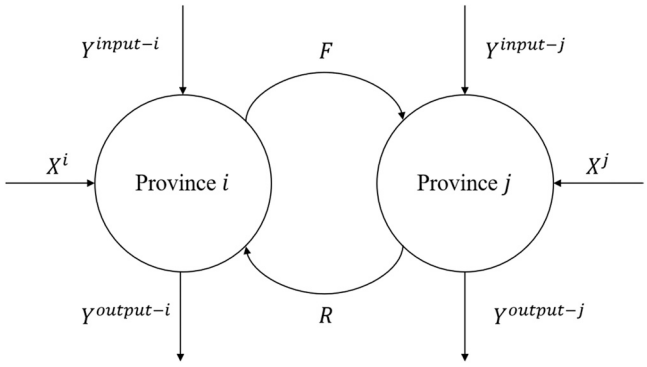


Fig. 3. Transport structure between two railway provinces.

$$\max(\omega_{s_0}^i * e_{s_0}^i - u_1 + \omega_{s_0}^j * e_{s_0}^j - u_2)$$

$$s.\ t. \begin{cases} \dfrac{U_i^T Y_s^{output-i} + Z^T F_s - u_1}{V_i^T(X_s^i,\ Y_s^{input-i}) + W^T R_s} \le 1 & \forall\ s \in S \quad \text{(i)} \\ \dfrac{U_j^T Y_s^{output-j} + W^T R_s - u_2}{V_j^T\left(X_s^j,\ Y_s^{input-j}\right) + Z^T F_s} \le 1 & \forall\ s \in S \quad \text{(ii)} \\ U_i^T,\ Z^T,\ V_i^T,\ W^T,\ U_j^T,\ V_j^T \ge \varepsilon & \text{(iii)} \\ u_1,\ u_2 \quad \textit{free in sign} & \text{(iv)} \end{cases} \tag{4}$$

With

$$e_{s_0}^i = \frac{U_i^T Y_{s_0}^{output-i} + Z^T F_{s_0} - u_1}{V_i^T(X_{s_0}^i,\ Y_{s_0}^{input-i}) + W^T R_{s_0}} \tag{5}$$

$$e_{s_0}^j = \frac{U_j^T Y_{s_0}^{output-j} + W^T R_{s_0} - u_2}{V_j^T(X_{s_0}^j,\ Y_{s_0}^{input-j}) + Z^T F_{s_0}} \tag{6}$$

where $\omega_{s_0}^i$ and $\omega_{s_0}^j$ are the weights of province i and province i under transportation line s_0; U_i^T, Z^T, V_i^T, W^T, U_j^T, V_j^T the weights of corresponding inputs and outputs variables; ε infinitesimal value (10^{-6}); u_1 and u_2 are free in sign variables for VRS. Specifically, when evaluating the transport service efficiency of an observed transportation line s_0, we define the province weights as the ratio of the inputs in the province to the total inputs of s_0, which is based on giving more weight to the provinces with high inputs. Thus, the weights defined in this way can effectively identify which province is more important in the transportation line. According to this definition, $\omega_{s_0}^i$ and $\omega_{s_0}^j$ are computed as

follows:

$$\omega_{s_0}^{i} = \frac{V_i^T(X_{s_0}^i, Y_{s_0}^{input-i}) + W^T R_{s_0}}{V_i^T(X_{s_0}^i, Y_{s_0}^{input-i}) + W^T R_{s_0} + V_j^T(X_{s_0}^j, Y_{s_0}^{input-j}) + Z^T F_{s_0}} \tag{7}$$

$$\omega_{s_0}^{j} = \frac{V_j^T(X_{s_0}^j, Y_{s_0}^{input-j}) + Z^T F_{s_0}}{V_i^T(X_{s_0}^i, Y_{s_0}^{input-i}) + W^T R_{s_0} + V_j^T(X_{s_0}^j, Y_{s_0}^{input-j}) + Z^T F_{s_0}} \tag{8}$$

such that objective function:

$$max \ \frac{U_i^T Y_{s_0}^{output-i} + Z^T F_{s_0} - u_1 + U_j^T Y_{s_0}^{output-j} + W^T R_{s_0} - u_2}{V_i^T(X_{s_0}^i, Y_{s_0}^{input-i}) + W^T R_{s_0} + V_j^T(X_{s_0}^j, Y_{s_0}^{input-j}) + Z^T F_{s_0}} \tag{9}$$

Following the same transformation step, the model (4) is equivalent to:

$$\max(U_i^T Y_{s_0}^{output-i} + Z^T F_{s_0} - u_1 + U_j^T Y_{s_0}^{output-j} + W^T R_{s_0} - u_2)$$

$$s.\ t. \begin{cases} U_i^T Y_s^{output-i} + Z^T F_s - u_1 - V_i^T(X_s^i, Y_s^{input-i}) \quad \forall\, s \in S & \text{(i)} \\ \quad - W^T R_s \le 0 & \\ U_j^T Y_s^{output-j} + W^T R_s - u_2 - V_j^T(X_s^j, Y_s^{input-j}) \quad \forall\, s \in S & \text{(ii)} \\ \quad - Z^T F_s \le 0 & \\ V_i^T(X_{s_0}^i, Y_{s_0}^{input-i}) + W^T R_{s_0} & \text{(iii)} \\ \quad + V_j^T(X_{s_0}^j, Y_{s_0}^{input-j}) & \\ \quad + Z^T F_{s_0} = 1 & \\ U_i^T, Z^T, V_i^T, W^T, U_j^T, V_j^T & \text{(iv)} \\ \quad \ge \varepsilon & \\ u_1, u_2 & \text{(v)} \\ \quad free\ in\ sign & \end{cases} \tag{10}$$

Then, we can obtain the transport service efficiency of the transportation line s_0 from model (11):

$$e_{s_0} = Z^T F_{s_0} + W^T R_{s_0} \tag{11}$$

where e_{s_0} is computed based on the transport service efficiency of the forward transportation line ($e_{s_i \longrightarrow j} = Z^T F_{s_0}$) and the transport service efficiency of the reverse transportation line ($e_{s_i \longleftarrow j} = W^T R_{s_0}$).

3.3. Potential improvement in exchange efficiency

In this section, we will discuss the potential improvement for the observed province in the railway supply chain network based on the MDEA-approach of Asmild et al. [1]. The advantage of the MDEA approach is that it can provide the target levels (and, thus, the maximum amount that can be reduced) per input, providing a direction for managers to improve efficiency (for MDEA applications, see [2,3,42]). The potential improvement (PI) is defined as:

$$PI = \beta^* = max\{\beta:\ d - \beta(d - d^*)\, \epsilon\ T\} \tag{12}$$

where d is an inefficient point which is inside the production possibility set (T); d^* is the ideal point. We firstly focus on the potential analysis of input improvements. Following linear programming models are formulated to calculate the ideal target levels (d^*) of every input $X(x_1, \ldots x_B)$ and imported goods (Y^{input}) in the observed province, respectively.

$$X_{b,0}^* = \min X$$

$$s.\ t. \begin{cases} \sum_{j=1}^n \lambda_j y_{p,j}^{output} \ge y_{p,0}^{output} & \forall\, p & \text{(i)} \\ \sum_{j=1}^n \lambda_j x_{b,j} \le X & b = 1,\ldots,B & \text{(ii)} \\ \sum_{j=1}^n \lambda_j x_{b,j} \le x_{b,0} & b = 1,\ldots,b-1,\ b+1,\ldots,B & \text{(iii)} \\ \sum_{j=1}^n \lambda_j y_{p,j}^{input} \le y_{p,0}^{input} & \forall\, p & \text{(iv)} \\ \sum_{j=1}^n \lambda_j = 1 & & \text{(v)} \\ \lambda_j \ge 0 & \forall\, j & \text{(vi)} \end{cases} \tag{13}$$

$$Y_{p,0}^{input*} = \min Y$$

$$s.\ t. \begin{cases} \sum_{j=1}^n \lambda_j y_{p,j}^{output} \ge y_{p,0}^{output} & \forall\, p & \text{(i)} \\ \sum_{j=1}^n \lambda_j x_{b,j} \le x_{b,i} & b = 1,\ldots,B & \text{(ii)} \\ \sum_{j=1}^n \lambda_j y_{p,j}^{input} \le Y & \forall\, p & \text{(iii)} \\ \sum_{j=1}^n \lambda_j y_{p,j}^{input} \le y_{p,0}^{input} & p = 1,\ldots,p-1,\ p+1,\ldots,P & \text{(iv)} \\ \sum_{j=1}^n \lambda_j = 1 & & \text{(v)} \\ \lambda_j \ge 0 & \forall\, j & \text{(vi)} \end{cases} \tag{14}$$

Then, the potential improvement index of the observed province in the railway supply chain network is expressed as:

$$\beta_0 = \max \beta$$

$$s.\ t. \begin{cases} \sum_{j=1}^n \lambda_j y_{p,j}^{output} \ge y_{p,0}^{output} & \forall\, p & \text{(i)} \\ \sum_{j=1}^n \lambda_j x_{b,j} \le x_{b,i} - \beta\left(x_{b,0} - X_{b,0}^*\right) & b = 1,\ldots,B & \text{(ii)} \\ \sum_{j=1}^n \lambda_j y_{p,j}^{input} \le y_{p,0}^{input} - \beta\left(y_{p,0}^{input} - Y_{p,0}^{input*}\right) & \forall\, p & \text{(iii)} \\ \sum_{j=1}^n \lambda_j = 1 & & \text{(iv)} \\ \lambda_j \ge 0 & \forall\, j & \text{(v)} \end{cases} \tag{15}$$

Next, the normalized potential improvement score (NPI) of each input of the observed province is calculated respectively as:

$$NPI_{b,0}^X = \frac{PI_{b,0}^X - min\{PI_{b,j}^X\}}{max\{PI_{b,j}^X\} - min\{PI_{b,j}^X\}},\ j = 1,\ \ldots,n,\ \ where\ PI_{b,j}^X = \beta_j(x_{b,j} - X_{b,j}^*) \tag{16}$$

$$NPI_{p,0}^{Y^{input}} = \frac{PI_{p,0}^{Y^{input}} - min\{PI_{p,j}^{Y^{input}}\}}{max\{PI_{p,j}^{Y^{input}}\} - min\{PI_{p,j}^{Y^{input}}\}},\ j = 1,\ \ldots,n,\ where\ PI_{p,j}^{Y_{input}} = \beta_j\left(y_{p,j}^{Input} - Y_{p,j}^{input^*}\right) \tag{17}$$

where the sub NPI-score of each input is between 0 and 1, with higher sub NPI-scores indicating more room for improvement in the respective input. Finally, the global NPI-score for the evaluated province is computed as:

$$NPI_0^{global} = \sum_b \omega_b{}^* NPI_{b,0}^X + \sum_p \omega_p{}^* NPI_{p,0}^{Y^{input}} \tag{18}$$

where ω_b and ω_p ($\sum_b \omega_b + \sum_p \omega_p = 1$) are the policy maker weights for the sub NPI-scores ($NPI_{b,0}^X$ and $NPI_{p,0}^{Y^{input}}$).

The potential analysis of output improvements is similar. The model can be constructed in the same idea as described above, which is not listed in the paper. We will discuss both input and output improvements based on the case study in the results.

4. An application to the Chinese railway

4.1. Data description

Based on the railway data in 2021, we study the railway supply chain exchange efficiency of 31 provinces of mainland China, excluding Hongkong, Macau, and Taiwan. As shown in Table 1, based on the discussion of exchange efficiency in Section 3.1, we select railway staff, length of transport route, and traffic inflow volume as input variables, and traffic outflow volume as output variable, respectively. The data of railway staff and length of transport route is collected from the *National Bureau of Statics*. The *China Railway Yearbook* provided us with the data of traffic inflow volume and outflow volume of each province.

The 31 provinces of mainland China are further divided into the following seven main regions: North China, Northeastern China, East China, Central of China, Southern China, Southwest China, and Northwestern China according to their geographical location in China as shown in Table 2.

4.2. Province exchange efficiency

Fig. 4 shows the exchange efficiencies of the 31 Chinese mainland provinces computed using the DEA-model as in model (2) and the input-output data for 2021. It can be seen that Shanxi, Tibet, Shanghai, and Hainan have the highest exchange efficiency among all 31 provinces. Shanxi's high exchange efficiency can be explained by the province's status of most cargo outbound provinces in China in combination with a relatively small cargo inflow volume. The reason behind it is that Shanxi is China's largest coal producer and exporter. According to the *2021 Shanxi Statistical Yearbook*, coal production in Shanxi accounted for a quarter of the country's coal production in 2021, and coal accounted for more than 90% of the province's railway cargo. Tibet's high exchange efficiency stems from its tiny railway workforce. With just 36 railway employees in 2021, Tibet had about 0.006 times the number of railway employees as Hainan, the province with the second-lowest workforce levels. The Tibet province also ranks second to last among all provinces in mainland China in terms of cargo export by rail and length of railway transport routes. Tibet's high exchange efficiency thus stems from its tiny railway workforce (which is low even for the limited cargo export and length of transport route). Shanghai is one of the most economically developed provinces in China, which has achieved a good balance between inputs and outputs, resulting in high exchange efficiency. For example, it employs relatively few staff, is not heavily on imports, and has high export volumes. Hainan is an island off the southern tip of China. Since the railway is not the main means of transportation in the province, it has a very low number of employees (second to last in the country) and a short length of transport route (first to last in the country) that account for its high exchange efficiency. The three provinces with the lowest exchange efficiency are Hunan, Heilongjiang, and Liaoning. For Hunan, the low exchange efficiency levels are due to a combination of a large number of railway employees and low traffic outflow volumes. Compared with Liaoning and Heilongjiang, Hunan province has fewer railway staff, but it has more natural resources and manufacturing enterprises, which means that the local government can prioritize more policies to stimulate exports to boost the local economy and balance the freight structure. Heilongjiang, in the northeast of the country, sees almost the same traffic inflow of railway cargo as the outflow. It has the highest number of railway employees among the 31 provinces but does not match this high number of railway employees with a sufficiently high freight volume in order to be exchange efficient. The relatively high workforce number of Heilongjiang is likely due to the fact that the bloated railway organizational structure in northeast China (compared with south China). In addition, the railway is the main mode of freight transportation in northeast China, and the railway infrastructure is well developed (with more stations, etc.), which further leads to a higher number of employees in this region than in other regions. Considering Heilongjiang's poor economic condition and poor product competitiveness, export volume is difficult to increase in the short term. The province can give priority to reducing the number of railway employees to improve exchange efficiency. Liaoning, also in the northeast, suffers from overstaffing and has the country's third-largest railway workforce. But unlike Heilongjiang, the volume of imports is 1.6 times that of exports, indicating that the province is heavily dependent on imports. Local governments could devise policies to encourage local production (for example, differentiating itself from the other provinces by producing products that are not or much less produced in the other provinces) to reduce dependence on other provinces.

Since the data of Shanxi and Tibet differ considerably from input-output data of the other Chinese provinces,[1] the average efficiency of the inefficient provinces as computed using model (2) is only 0.29. In order to further understand the railway supply chain exchange efficiency of other provinces and to put forward more practical suggestions (i.e., improvement paths) to decision-makers, we use the multi-level DEA method[2] to carry out a robust analysis of the exchange efficiencies of the Chinese provinces in 2021 which is shown in Fig. 5. By continuously removing efficient provinces and re-applying the model (2), a total of 7 railway supply chain exchange efficiency frontiers are obtained. It can be seen that the provinces that constitute higher-level efficiency frontier are mostly located in the north and west of China, with a few in the east and south. South-central provinces such as Sichuan, Hubei, Hunan, Jiangxi, Yunnan, Guangxi, and Anhui lie on the lower-level efficiency frontiers. In order to improve the exchange efficiency more realistically, it is recommended that the provinces located on the lower-level efficiency frontiers learn from the provinces that are located on the exchange efficiency frontiers only one or two levels above them. Table 3 shows that the inefficient provinces have an average exchange efficiency of 0.29 and 0.51 under 1st and 2nd level exchange efficiency frontiers, which means the inefficient provinces are far away from these higher-level efficiency frontiers and the provinces that constitute these frontiers in terms of exchange efficiency (i.e., Shanxi, Tibet, Shanghai and Hainan locate on the 1st level exchange efficiency frontier, Tianjin, Shaanxi, Ningxia and Inner Mongolia locate on the 2nd level exchange efficiency frontier). Table 3 also clearly shows that after the 3rd level the changes in average exchange efficiency among remaining inefficient provinces becomes reduced. This shows that the exchange efficiencies of the provinces that are positioned below the 3rd level exchange efficiency frontier (4th level exchange efficiency frontier to 7th level exchange efficiency frontier) are positioned more closely to each other in terms of input and output data and in exchange efficiency.

4.3. Transportation line transport service efficiency

Fig. 6 reports the ranking of the transport service efficiency of the transportation lines between provinces. In Fig. 6(a), we combine the transport service efficiency values (obtained from model (11)) of all provincial transportation lines to obtain a region-level transport service efficiency score (by adding the transport service efficiency scores of the

[1] Shanxi is the largest coal producer and exporter in China as mentioned in the result of Fig. 4. In 2021, its traffic outflow volume was half of the total number of other provinces or 17 times the average of other provinces. The data for Tibet is illustrated in the results of Fig. 4.

[2] The DMU inside the frontier sometimes needs to reduce inputs or increase outputs a lot to reach the frontier, which usually cannot be accomplished in a period of time. Therefore, the multi-level DEA method provides an approach to gradually improve the efficiency of inefficient DMUs by establishing multiple efficient frontiers. It is also a good tool for robustness testing [15,27]. In this paper, after getting the result of Fig. 4, we eliminate the efficient DMUs and apply the model (4) again to obtain the new frontier. Repeat this step to get multiple frontiers.

Table 1
Variables of railway of 31 Chinese provinces in 2021.

Variable	Unit	Mean	Minimum	Maximum	Standard deviation
Railway staff	Person	59,155	36	126,389	34,980
length of transport route	10,000 kilometers	0.42	0.05	1.28	0.24
Traffic inflow volume	10,000 ton	6962	117	61,472	10,648
Traffic outflow volume	10,000 ton	6962	42	78,132	14,054

Table 2
Seven regions and corresponding provinces.

Region	Provinces
North China	Beijing, Tianjin, Hebei, Shanxi, Inner Mongolia
Northeastern China	Heilongjiang, Jilin, Liaoning
East China	Shanghai, Jiangsu, Zhejiang, Anhui, Fujian, Jiangxi, Shandong
Central of China	Henan, Hubei, Hunan
Southern China	Guangdong, Guangxi, Hainan
Southwest China	Chongqing, Sichuan, Guizhou, Yunnan, Tibet
Northwestern China	Shaanxi, Gansu, Qinghai, Ningxia, Xinjiang

transportation lines that are located in the region). For example, the transport service efficiency for the North-East regions is derived from the transport service efficiency scores of all province-to-province transportation lines that are located in these two regions). It is shown clearly that North China, Northeastern China, East China, and Southwestern China have higher intra-regional transport service efficiency compared with inter-regional transport service efficiency. Combined with the geographical location in Fig. 4, we can find that the transport service efficiency ranking of two adjacent regions is usually higher than that of two regions with a longer spatial distance, which shows that the transport service efficiency is closely related to the distance. For example, the transportation lines from Northeastern to East China, North to Southern China, and Northeastern to Southern China are doing worst in terms of transport service efficiency. In addition, it is worth noticing that the transport service efficiency of the transportation lines from Northeastern China to the other regions in China (except Northeastern to North China, and inner Northeastern) are among the lowest observed (with transportation lines from Northeastern to Southern China ranking the lowest in transport service efficiency). This poor transport service efficiency of the transportation lines connecting Northeastern China to (most of) the other regions in China may be due to the weak economy of Northeastern China. Another



Fig. 4. Exchange efficiencies of the Chinese provinces (year 2021).



Fig. 5. Exchange efficiencies of Chinese provinces at different levels.

Table 3
The average exchange efficiency of inefficient provinces at each level.

Level	1st	2nd	3rd	4th	5th	6th	7th
AEEIP	0.29	0.51	0.69	0.73	0.84	0.91	1

explanation is that the three northeastern provinces (Heilongjiang, Jilin and Liaoning) are generally less connected by railway supply chain with provinces in the other regions.

Fig. 6(b) shows the ranking of transport service efficiencies of all province-to-province transportation lines. Among the 465 transportation lines, Shanxi-Hebei in North China has the highest transport service efficiency. It is worth noting that, among all the transportation lines, there are 11 lines with zero transport service efficiency scores due to zero traffic outflow volume between two stations that are connected by these transportation lines. 10 of these transportation lines are from Hainan to other provinces (i.e., Tianjin, Jilin, Heilongjiang, Fujian, Shandong, Tibet, Shaanxi, Gansu, Qinghai, and Ningxia). In combination with the earlier finding of a high exchange efficiency of the province Hainan as shown in Fig. 4, Fig. 6(b) provides a way for Hainan local government to improve its provincial exchange efficiency further to distinguish it from other efficient provinces. In particular, one way for Hainan to improve the exchange efficiency would be to start railway freight exchange with the above-mentioned provinces. Another interesting point is that provinces with low exchange efficiency scores do not necessarily have only transportation lines with low transport service efficiencies. In fact, it is perfectly possible for a province with low exchange efficiency to have one or a few transportation lines that score high in terms of transport service efficiency. An example of such a province is Hunan which has the third-lowest exchange efficiency but has some transportation lines (example, the transportation line connecting Hunan to Jiangxi) that ranks high (11th) in transport service efficiency.

4.4. Forward and reverse line transport service efficiency

Fig. 7 shows the average ranking of the transport service efficiencies of the forward and reverse transportation lines between regions (Fig. 7(a)) and between provinces (Fig. 7(b)). In the ranking of the transport service efficiencies of the transportation lines averaged between the regions as in Fig. 7(a), it can be seen that the transport service efficiency of transportation lines from East, Central, and Southern China to North and Northeastern China are significantly lower than the transport service efficiency of the transportation lines in the opposite direction. A possible reason for this result is due to the central government's strategy of moving energy resources from North and Northeastern to East, Central and Southern China. That is, the economy in North China is generally rich in heavy industry due to the wide distribution of energy resources such as coal and steel, but rather weak in light industry. Provinces in South China can more easily import these light industry products from surrounding provinces in Central rather than from provinces in North China. On the other hand, provinces in the north of China have little import from South due to economic stagnation and cost of long-distance transportation. The above finding further reveals the reasons for the low transport service efficiency of transportation line from Northeastern to Southern China in Fig. 4(a). In addition, Fig. 7(a) indicates that the transport service efficiency of transportation lines from Central to Southern China is close to the transport service efficiency of the transportation lines in the opposite direction. This may be due to the relative balance of imports and exports in the southern and central provinces with comparable economic strength. Thus, based on the above, the transport service efficiency of railway transportation lines

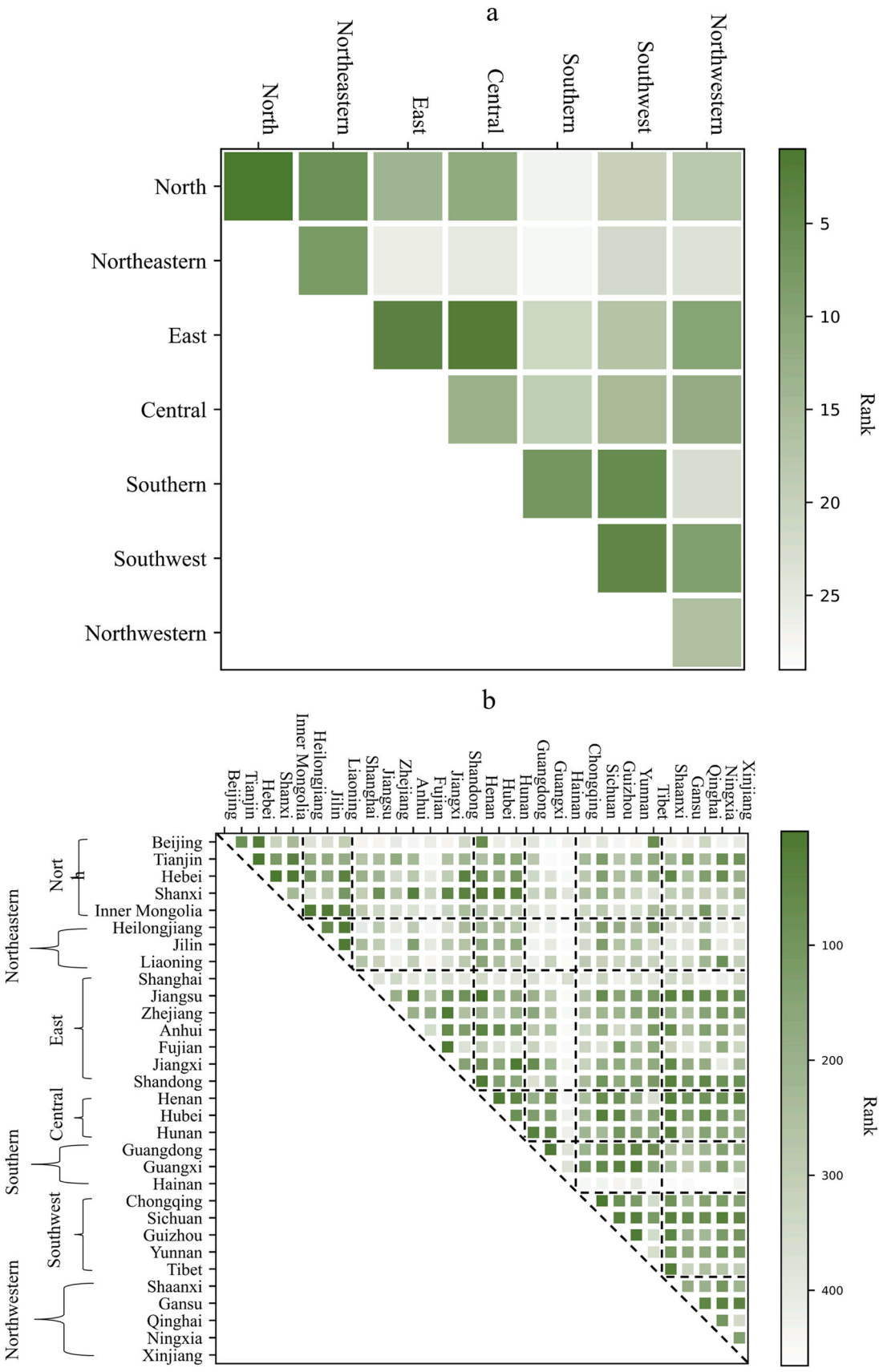


Fig. 6. Transport service efficiency ranking of the transportation lines per province and per region.

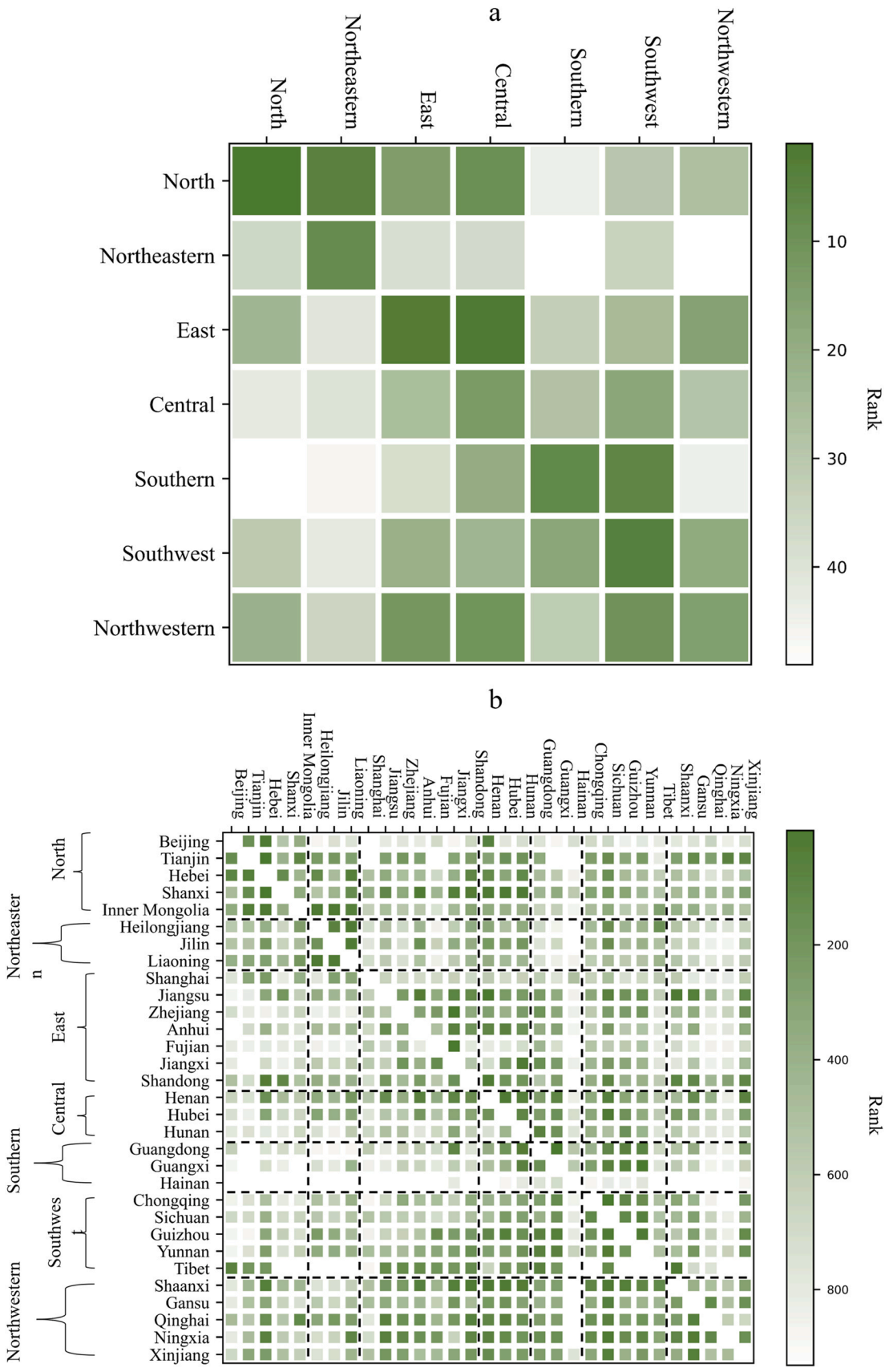


Fig. 7. Transport service efficiency ranking of the forward and reverse lines of transportation lines per province and per region.

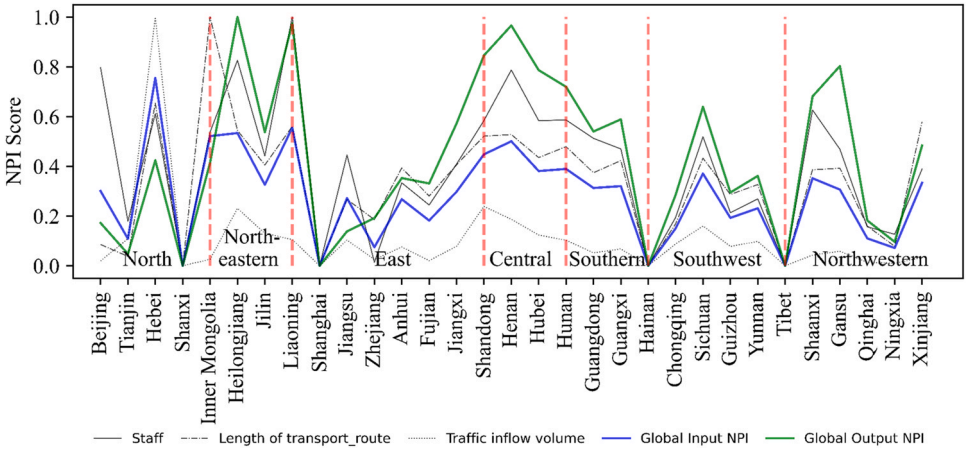


Fig. 8. Potential improvement score of Chinese provinces.

from North to South China is generally higher than the transport service efficiency from the transportation lines in the opposite direction. Considering China's east-west direction, the transportation lines from East to Central China, Western (Northwestern and Southwestern) to Central China are generally more transport service efficient than the transportation lines in the opposite direction. One possible reason is that due to the low Gross Domestic Product (GDP), provinces in West China have lower purchasing ability (thus importing less from provinces in the other parts of China). However, provinces in western China typically have abundant agricultural products that can be exported. This results in more exported commodities than imported commodities. The eastern coastal provinces have a large amount of seaborne goods transferred to the central region by rail.

Fig. 7(b) reports more details about the transport service efficiency ranking of all forward and reverse railway transportation lines at the provincial level which is essentially the breakdown of Fig. 7(a). It can help local government in analyzing the result of Fig. 6(b). First, although Shanxi-Hebei has the highest transport service efficiency, it can be found that the transport service efficiency of the railway transportation line from Shanxi to Hebei is highest among all 930 transportation lines, while the transport service efficiency of the transportation line from Hebei to Shanxi ranks 133th. The above results indicate that there is a huge difference in transport service efficiency of the transportation line depending on the direction. A total of 50 forward and reverse transportation lines are evaluated with transport service efficiency score of zero due to no data and/or missing data, which is more than we would expect given that there were 10 transportation lines with zero transport service efficiency scores equal to zero in Fig. 6(b). The reason is that some provinces only have one-way cargos exchange, which makes that the transportation line obtains a non-zero transport service efficiency score but one of the two composing transportation lines (forward or reverse line) does obtain a zero transport service efficiency score (due to no cargo exchange in that direction). Such detailed information on the transport service efficiency of transportation lines gives managers more detailed insights on how to improve provincial exchange efficiency. For example, expanding two-way transportation lines may be easier and better in terms of transport service efficiency than expanding no-exchange transportation lines because rail infrastructure is already in place between the two provinces, and both sides are more familiar with each other's transport policies so that it is less expensive to conclude trade agreements.

4.5. Potential improvements of inputs and outputs

Fig. 8 shows the global input and output *NPI*-score (calculated based on the sub *NPI*-scores of the different input of model (18) thereby using equal weighting for the inputs for global input *NPI*-score, the same approach to the global output *NPI*-score) and its breakdown into sub *NPI*-score per input and output for each of the provinces. It is obvious that Hebei has the highest global input *NPI*-score (0.76). Its traffic inflow volume *NPI*-score equals 1 which indicates that Hebei is the province that relies most on the import of cargo by railway supply chain among the 31 provinces. The traffic inflow volume *NPI*-scores in Fig. 8 also indicate that Heilongjiang, Shandong, and Sichuan have plenty of room for improvement in their balance between rail imports and exports. As for the input 'Staff', the sub *NPI*-scores show that Heilongjiang, Liaoning, Beijing, and Henan have more employees than other provinces. Thus, for these provinces one possible way to improve the efficiency would be to downsizing the railway staff. For the length of transport route, the sub *NPI*-scores demonstrate that Inner Mongolia, Hebei and Xinjiang constitute the top three. The possible reason why for Hebei the sub *NPI*-score for the length of transport route is higher is that more transportation routes are used for passenger transportation and fewer transportation lines are used for railway freight transportation. The high length of transport route *NPI*-scores of Inner Mongolia and Xinjiang are due to the mismatch between their mileage and the amount of cargo transported. Inner Mongolia and Xinjiang have large areas and long railway transportation lines, but the relatively poor economy and the amount of cargo exchanged is relatively small. As shown in Fig. 4, among the 31 provinces, Shanxi, Shanghai, Hainan and Tibet are the most efficient, so their global input and output *NPI*-score is 0 in Fig. 8. The global output *NPI*-scores (since there is only one output variable in Table 1, the global output *NPI*-score here is the score of traffic outflow volume) show that Liaoning, Heilongjiang, and Henan have the highest potential to increase their traffic outflow volume respectively. Local governments in all three provinces should prioritize stimulating exports to improve exchange efficiency. In particular, provinces in northeast, central and southern regions have both high input and output *NPI*-scores. Governments in these regions could consider a mix of inputs and outputs policies for improvement.

5. Conclusion

This paper analyzed the railway supply chain performance of Chinese provinces in 2021 by using a novel concept of exchange efficiency via the proposed DEA model. To obtain more detailed insights on provincial exchange efficiency, a new two-stage method was proposed to measure the transport service efficiency of the forward and reverse railway transportation lines between provinces. In addition, reginal transport service efficiency is analyzed to provide policy makers with a high-level view of China's railway supply chain. Finally, the MDEA method is used to evaluate the possible input reduction of each province, which provides the improvement path for decision-makers to improve provincial exchange efficiency.

The results provide the central government with a whole picture of the railway freight exchange efficiency of China's provinces in 2021. Among 31 provinces, Shanxi, Tibet, Shanghai, and Hainan were evaluated as being most exchange efficient. The transport service efficiency of intra-regional rail transport (thus between provinces in the same region such as North, East and Southwestern China) was found to be usually higher than the transport service efficiency of cross-regional rail transport (thus between provinces of different regions). Results also showed that the transport service efficiency of railway transportation lines from the North to the South is more efficient than the transportation lines in the opposite direction. Since transport service efficiency is related to the total volume of railway freight imports and exports in each province, the above results not only provide decision-makers and government with insights into the transport service efficiency of railway transportation lines in the Chinese regions and provinces but also a general idea on the transport service efficiency in transportation of goods by railway supply chain in China. This provides the basis for the government to adjust the freight structure. Lastly, the potential efficiency improvement analysis provides *NPI* scores for each input and output of the province to help decision-makers improve exchange efficiency.

Based on the research results, in order to improve efficiency, we suggest that regions with low railway transport efficiency in China, especially those with high import dependence, should prioritize improving the economy, adjusting the industrial structure, and balancing the proportion of goods imports and exports within the region, rather than reducing the fixed investment in the railway. In addition, the provinces located on the lower-level efficiency frontiers should learn from the provinces that are located on the exchange efficiency frontiers only one or two levels above them to improve efficiency.

This study has the following limitations. First, not only do we lack data on rail cargo in Hong Kong, Macao, and Taiwan, but the existing data cannot show the specific type of cargo. Second, we do not distinguish between the high-speed railway supply chain and general speed railway supply chain between two provinces, missing the breakdown of transport service efficiency between different train types. Third, only rail transport is considered in this study, whereas in national transport systems (such as in China), road and sea transport may carry more volumes. Therefore, the exchange efficiency of multimodal transport is of great significance in future research.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The paper has been supported by the China Scholarship Council (202008310250), Shanghai Maritime University (2021YBR016).

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